

# Answer Key — Biology Grade 11

Sub-Strand 2.3: Excretion in Plants • Lesson 1: Where Do Plant Wastes Go? Gum, Droplets and Falling Leaves

Teacher reference only. Do not project or hand out.

## Quick marking grid

| Q1 | Q2 | Q3 | Q4 | Q5 | Q6 | Q7 |
|----|----|----|----|----|----|----|
| D  | A  | B  | A  | C  | D  | B  |

USE AFTER: PREDICT PHASE — Nature Walk: droplets on the edges of grass and arrowroot leaves

**Q1. On a humid early morning, Grade 11 learners looked closely at grass and arrowroot (taro) leaves in the school garden. What did they see standing along the edges of the leaves?**

- A. Yellow leaves dropping off
- B. Thick milky sap oozing out
- C. Amber resin that had hardened into a seal
- D. Small droplets of liquid**

**Correct answer: D.** Droplets of liquid along the leaf edges were the observation on grass and arrowroot leaves. Milky sap and resin came from cut Acacia or Euphorbia stems, and yellow leaves were seen fallen under trees.

USE AFTER: PREDICT PHASE — Think-Pair-Share of the comparative diagrams and the prediction 'I think a plant gets rid of its wastes by...'

**Q2. During Think-Pair-Share, a learner predicts: 'The sticky sap from a cut Acacia is the plant's blood.' At this point in the lesson, what is the best scientific way to treat this idea?**

- A. Record it as a prediction and test it later using evidence**
- B. Reject it at once because the teacher has not agreed with it
- C. Ignore it because predictions made before teaching do not matter
- D. Accept it as true because a classmate said it

**Correct answer: A.** A prediction is an idea that can be tested and revised. Predict-Observe-Explain asks learners to commit to an idea first and check it with evidence later.

USE AFTER: OBSERVE PHASE — Step 2: annotating the chart comparing plant and animal metabolic rates

**Q3. A chart compares a running goat with a maize plant of the same body mass. It shows that the goat uses much more oxygen and food and makes much more nitrogenous waste. What does the chart show about the amount of nitrogenous waste the maize plant makes?**

- A. The same amount as the goat
- B. Much less than the goat**
- C. None at all
- D. More than the goat

**Correct answer: B.** The chart shows that plants have a much lower metabolic rate and make much less nitrogenous waste per kilogram than animals. It does not show that plants make none.

USE AFTER: OBSERVE PHASE — Step 1 brainstorm on metabolic wastes and the introduction of the term excretion (excretion versus egestion)

**Q4. Excretion is the removal of metabolic wastes from the body. Which of these is an example of excretion in a human?**

- A. Removing urea in urine**
- B. Breathing in oxygen
- C. Swallowing food
- D. Passing out undigested ugali fibre as faeces

**Correct answer: A.** Urea is a metabolic waste made inside body cells, so removing it in urine is excretion. Undigested fibre was never part of metabolism, so its removal is egestion, not excretion.

USE AFTER: EXPLAIN PHASE — Structured debate 'Plants do not need kidneys' and the gaps still unexplained, such as the leaf-edge droplets

**Q5. After the debate, a learner says: 'We now have ideas about why plants have no kidneys, but we still cannot say what the droplets on the edges of arrowroot leaves are or how they form.' Which statement is correct at this stage of the unit?**

- A. The plant is sick and is leaking liquid
- B. The droplets are sweat from sweat glands on the leaf
- C. How the droplets form has not been explained yet and will be investigated in the next lesson**
- D. The droplets are urine made by tiny kidneys inside the leaf

**Correct answer: C.** The lesson deliberately leaves the leaf-edge droplets unexplained. Learners name this as a gap to investigate in Lesson 2, and the other choices are untested misconceptions.

USE AFTER: DQB PHASE — Driving Question Board: writing and clustering sticky-note questions

**Q6. Learners are writing sticky-note questions for the Driving Question Board about oozing sap, leaf-edge droplets and falling leaves. Which question best fits the board, because it is about the phenomenon and cannot be answered yet?**

- A. Did we go on a nature walk today?
- B. How many leaves does one arrowroot plant have?
- C. What colour is the gum from an Acacia?
- D. Why do arrowroot leaves have water drops along their edges in the morning?**

**Correct answer: D.** A good DQB question starts with How or Why and asks about the phenomenon, and learners cannot yet answer it. The other questions can be answered by looking, counting or recalling.

USE AFTER: MODEL BUILDING PHASE — Revising the Day 1 comparative diagram with a different coloured pen and marking '?' on routes not yet explained

**Q7. While revising her drawing, a learner puts a question mark on the arrow showing how wastes leave the plant, because she cannot explain it yet. Why is this a good scientific move?**

- A. It means she did not take part in the lesson
- B. It shows a gap in her model that needs to be investigated**
- C. It shows the drawing is completely wrong and must be rubbed out
- D. It means the teacher will give the answer without any investigation

**Correct answer: B.** Marking unexplained routes with a '?' turns gaps into questions for investigation. It makes the growth in her thinking visible, and it prepares the next lesson's puzzle.